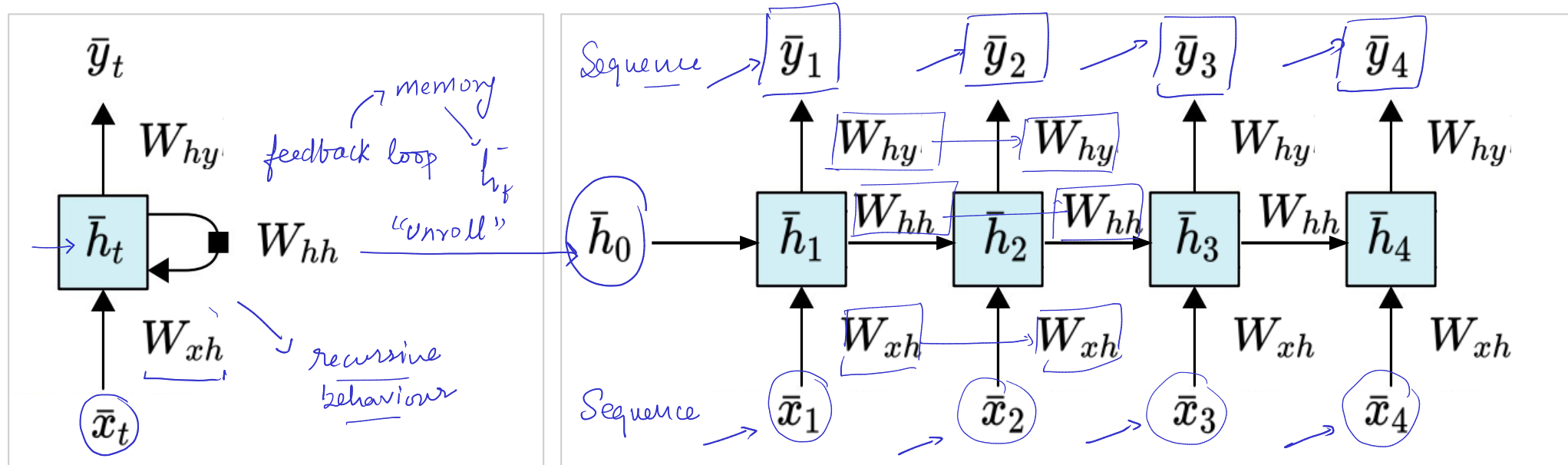


A simple recurrent neural network



$$\bar{h}_t = f(\bar{h}_{t-1}, \bar{x}_t)$$

memory state $(t) \equiv \text{Dep}(\text{MS}(t-1), \text{current I/P})$

$$\bar{h}_t = \tanh(W_{xh}\bar{x}_t + W_{hh}\bar{h}_{t-1})$$

$$\bar{y}_t = W_{hy}\bar{h}_t$$

$$\bar{y}_t = F_t(\bar{x}_1, \bar{x}_2, \dots, \bar{x}_t)$$

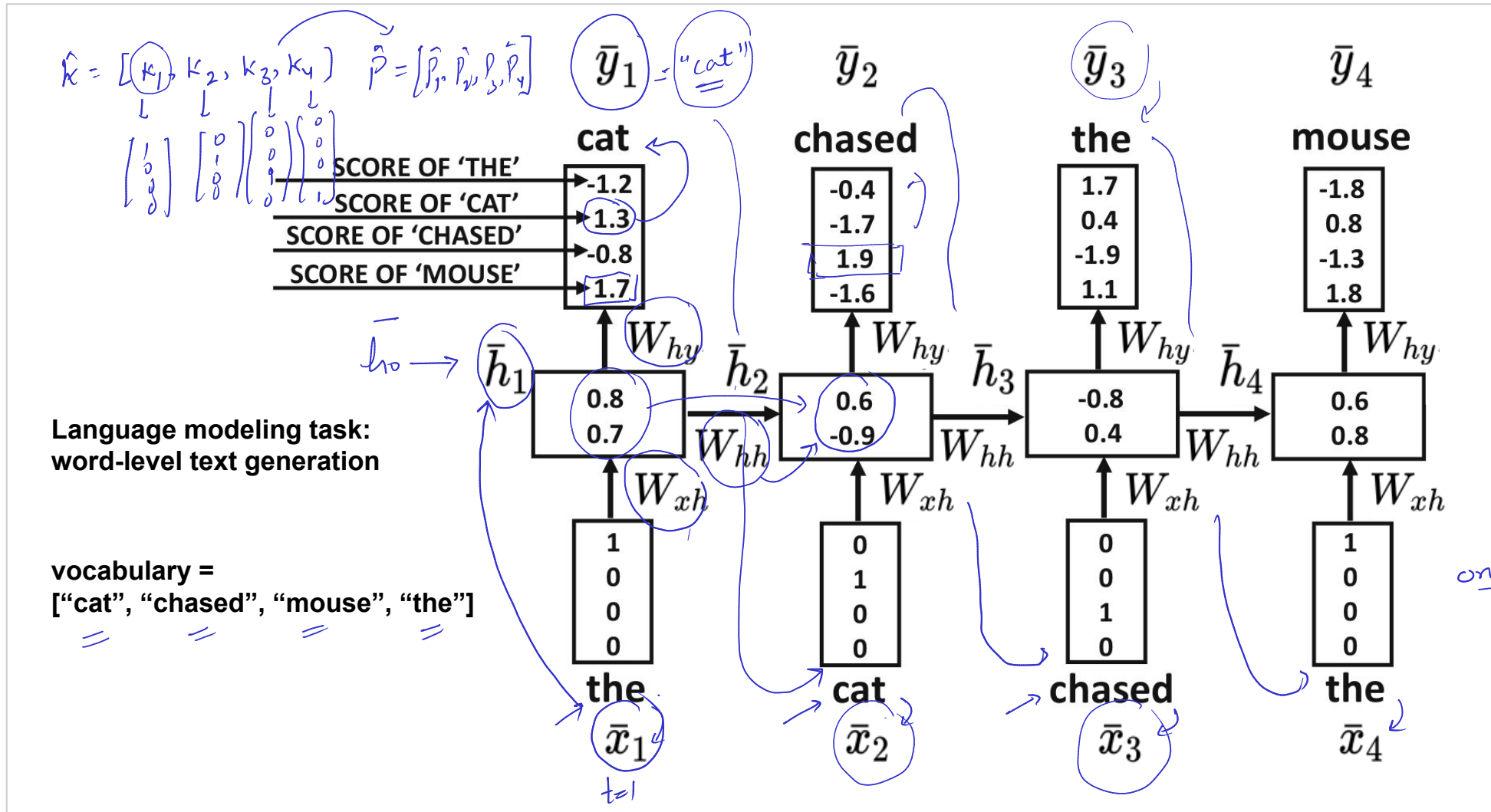
$\downarrow$
RNN

- Same weights are used across all timesteps.
- Nonlinearities are induced using activations.

$$\bar{h}_2 = f(\underbrace{f(\bar{h}_0, \bar{x}_1)}_{\bar{h}_1}, \bar{x}_2)$$

A simple example: language modeling

$$\bigcirc \rightarrow \bigcirc \rightarrow \bigcirc \rightarrow \bigcirc$$



one-hot encoding representation

